ENEL 573 Computer Networks: HW Assignment 4 (Fall 2026)

(Due on Nov 6th, at 11:59pm Mountain Time)

Name: _____ UCID: _____ Score: _____

Problem 1 (10 points):
An organization with a class B address needs at least 950 subnets. Find the subnet mask and the configuration of each subnetwork.

Problem 2 (10 points):
Find the subnetwork address and the host id for the following:
IP address: 120.14.122.16
Mask: 255.255.192.0

Problem 3 (10 points):
Consider sending a 1300-byte datagram into a link that has a MTU of 500 bytes. Which of the following is true? Please justify your answer.
1. Three fragments are created with offsets 0, 500/8 1000/8
2. Three fragments are created with offsets 0, 480/8 and 960/8.
3. Three fragments are created, with offsets 0, 460/8 and 920/8.
4. None of the above

Problem 4 (10 points):
Suppose an IP datagram is fragmented into 10 fragments, each with an independent loss probability of 1%. What is the probability of receiving the datagram correctly at the destination if (a) the datagram (each fragment) is transmitted once; (b) the datagram (each fragment) is transmitted twice?

Problem 5 (10 points):
Suppose nodes A, B, and C each attach to the same broadcast LAN (through their adapters). If A sends thousands of IP datagrams to B with each encapsulating frame addressed to the MAC address of B, will C's adapter process these frames? If so, will C's adapter pass the IP datagrams in these frames to the network layer C? How would your answers change if A sends frames with the MAC broadcast address (i.e., ARP request frames)? Please justify your answers.